Ostad

AI-Driven Cyber Security & Ethical Hacking Career Track Program

Batch-12

ASSIGNMENT REPORT ON

Research on AI in Cybersecurity & Cyber Law

Submitted By:	Muhammad Abu Hurayra
Submitted To:	Sheikh Mohammad Adil
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Task-1: Research on AI in Cybersecurity

Part 1: AI in Cybersecurity — The Defense & The Offensive

AI acts like a 24/7 smart security guard. Instead of relying solely on static rules, AI algorithms continuously monitor network traffic, identify anomalous behavior, and automatically neutralize threats in real time.

Key AI-Powered Hacking & Pentesting Tools

PentestGPT: An AI assistant that guides ethical hackers step-by-step through discovering and exploiting security loopholes.

XBOW: An automated AI tool designed to autonomously scan systems for bugs and vulnerabilities.

Burp AI: An intelligent feature integrated into Burp Suite that drastically speeds up web traffic analysis and flaw detection.

The Pros and Cons of Security AI

Advantages: Lightning-fast data analysis, round-the-clock protection without fatigue, and the ability to detect unknown (zero-day) threats based on behavior patterns.

Limitations: Prone to false alarms (flagging benign files as threats), lacks human intuition/common sense, and can be weaponized by malicious actors to construct sophisticated attacks.

The Future Trend: Cyberspace is moving toward an “AI vs. AI” battleground — where offensive AI-driven exploits will meet defensive AI engines capable of self-healing vulnerabilities before exploitation.

Task-2: Research on Cyber Law & Ethics in the AI Era

1. What is Cyber Law

Cyber Law is a set of legal rules that controls and regulates everything on the internet. It protects users from cyber crimes like hacking, online fraud, or data theft, and defines penalties for breaking these rules.

2. What is Cyber Ethics

Cyber Ethics refers to the moral principles and good behavior we should maintain while using the internet. It is not a legal law, but rather user responsibility—like respecting others' privacy, not spreading fake news, and avoiding plagiarism.

3. Cybercrime in the AI Era

With AI, cybercriminals have become more dangerous. They are now using AI to create convincing deepfakes for blackmailing, generating automated phishing emails that look 100% real, and launching smart malware attacks that can bypass basic firewalls.

4. Bangladesh Cyber Security Act (Basic Overview)

The Cyber Security Act (CSA) of Bangladesh is the legal framework designed to combat digital crimes and protect the country's cyberspace. It punishes activities like hacking, unauthorized system access, spreading hate speech, and online harassment with fines and imprisonment.

5. Legal Use of AI in Cybersecurity

Using AI legally means using it strictly for defense and within authorized boundaries. Security teams use AI to monitor network traffic, identify vulnerabilities, and automatically block malicious activities with proper organization approvals.

6. Ethical Use of AI Tools

Ethical use means using AI responsibly without harming others. This includes avoiding the creation of fake content, respecting user privacy, and ensuring that AI tools

(like PentestGPT or ChatGPT) are only used for learning or authorized penetration testing.

7. Rules of Engagement (ROE) and Scope

Rules of Engagement (ROE): A formal agreement that defines the exact rules, limitations, and permissions before conducting a security test on a system.

Scope: The clear boundary specifying exactly which assets, servers, or networks are allowed to be tested, and which ones must not be touched.

8. Importance of Responsible Cybersecurity Practices

Cybersecurity is not just about tools; it is about responsibility. Practicing it ethically ensures that user privacy is protected, systems remain stable, and digital environments stay safe for everyone. Without responsibility, security tools can easily turn into dangerous weapons.

What I Have Learned

From this assignment, I have gained a solid understanding of how Artificial Intelligence is transforming both the technical and legal sides of cybersecurity. In Task-1, I learned about advanced AI tools like PentestGPT and XBOW, and how they assist in defense and vulnerability scanning. In Task-2, I discovered the critical role of Cyber Law, Cyber Ethics, and the Bangladesh Cyber Security Act (CSA) in maintaining digital safety. Overall, I now understand that while AI is a powerful tool for stopping modern threats, it must be used responsibly, ethically, and within a strictly defined legal scope to keep our cyberspace safe.

Sources / References

Ostad AI Academy Course Materials & Live Classes

Google Search & Official Tech Documentations

TryHackMe & OWASP Cybersecurity Resources

Bangladesh Cyber Security Act (CSA) Official Overview

PortSwigger Web Security Academy & Cybersecurity Blogs